

2024 Edition

SIGNAL SURGE

The Annual Magazine of
SSN IEEE Signal Processing Society
Student Branch

ssn



About us

Founded as IEEE's first society in 1948, the Signal Processing Society is the world's premier association for signal processing engineers and industry professionals. Engineers around the world look to the Society for information on the latest developments in the signal processing field. Its deeply rooted history spans 75 years, featuring a membership base of more than 20,000 deeply interested and involved signal processing engineers, academics, industry professionals and students who are all part of a dynamic global community – spanning 120 countries worldwide

To advance the knowledge of signal processing and extend the benefits of the Signal Processing Society, the SSN IEEE SPS Student Branch was set up on August 21, 2023. As we mark the first anniversary of our student branch, we wanted to showcase our achievements to the wider community, which led to the inception of our annual magazine.

"Signal Surge" highlights the workshops, events, and accomplishments of our members. It serves as a testament to the hard work and dedication of our student branch. In this magazine, you will find detailed accounts of the collaborative events that brought together experts and enthusiasts, and the numerous awards and recognitions our members have received.

We believe that "Signal Surge" will be a great source of inspiration and motivation for all signal processing enthusiasts at SSN. Our goal is to foster a vibrant community of learners and innovators who are passionate about the potential of signal processing to transform the world.

Through this publication, we aim to not only celebrate our achievements but also to encourage others to join us in exploring the exciting possibilities within the field of signal processing. Welcome to the inaugural edition of "Signal Surge" – a reflection of our journey and a glimpse into the promising future of signal processing at SSN.

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ABOUT OUR COLLEGE

SSN College of Engineering(SSNCE) was established in 1996 with the aim of providing quality higher education and adopts innovative methods to improve the quality of education on a consistent basis. The campus has a cosmopolitan atmosphere with combination of some of the finest graduate, undergraduate, and research programs, enhanced by the most skillful and experienced faculties, remarkable facilities, and a verdant landscape spanning a vast expanse of 230 acres amidst serene surroundings. Experienced and learned professionals strongly nurture the students. The global standards set at SSN in the field of teaching and research spur us on in our relentless pursuit of excellence. In fact, it has become a way of life for us. The highly motivated youngsters on our campus are a constant source of pride. Founded by the visionary Mr. Shiv Nadar, who serves as the illustrious Chairman Emeritus and guiding force behind HCL Technologies, notable to promote, conceptualize and bring about a paradigm shift through the development of outstanding leadership, research, knowledge and ideas for education and allied development sectors.





Institute of Electrical and Electronics Engineers, is the world's largest professional organization which aims to use advanced technology for assisting human needs. Established in 1963 through the merger of the American Institute of Electrical Engineers (AIEE) and the Institute of Radio Engineers (IRE).

IEEE today has

- Over 460,000 members in more than 190 countries, with more than 66 percent from outside the United States
- More than 171,000 Student members
- 344 Sections in ten geographic Regions worldwide
- 2,709 Chapters that unite local members with similar technical interests
- 3,635 Student Branches at colleges and universities in over 100 countries
- 4,194 Student Branch Chapters of IEEE technical Societies

In addition to fostering member collaboration and knowledge exchange, IEEE sets industry standards for innovation and advances the development and application of technology to address societal concerns. IEEE is a major force behind technological advancement and the direction of engineering and technology around the world through conferences, publications, and educational activities.

Mission :

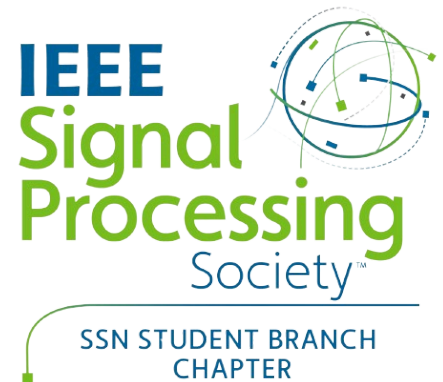
IEEE's core purpose is to foster technological innovation and excellence for the benefit of humanity.

Vision :

IEEE will be essential to the global technical community and to technical professionals everywhere, and be universally recognized for the contributions of technology and of technical professionals in improving global conditions.

ABOUT IEEE SPS

Founded as IEEE's first society in 1948, the Signal Processing Society is the world's premier association for signal processing engineers and industry professionals. Engineers around the world look to the Society for information on the latest developments in the signal processing field. Its deeply rooted history spans 75 years, featuring



a membership base of more than 20,000 deeply interested and involved signal processing engineers, academics, industry professionals and students who are all part of a dynamic global community – spanning 120 countries worldwide. The Society organizes numerous conferences around the world every year, focusing on the innovations shaping the future of signal processing and the future of our world. Some of the benefits of joining IEEE SPS are:

1. Signal Processing Repository (SigPort) is an online repository of manuscripts, reports, technical white papers, theses, and supporting materials such as videos.
2. Exclusive Discounts: Savings on conference registrations, SigPort uploads, SPS publications, and upcoming PDH/CEU credits.
3. Global Networking: Opportunities to network with leaders, join communities, and connect through local and technical events.
4. Volunteer Opportunities: Engage in society activities, choose your involvement level, and contribute through blogs, boards, or organizing events.
5. The IEEE Signal Processing Society (SPS) Education Program provides a wealth of resources, knowledge, and programs across various signal processing disciplines. These offerings include Digital Resources, Live In-Person Programming, Diverse Educational Formats, Global Expertise.
6. The IEEE Signal Processing Society (SPS) offers several scholarship and award opportunities, including the IEEE Signal Processing Society Travel Grants, IEEE SPS Student Paper Contest, IEEE SPS Summer School Scholarships, IEEE SPS Distinguished Lecturer Program, IEEE SPS Four-Year PhD Fellowship, IEEE SPS Regional Student Paper Awards, and IEEE SPS Undergraduate Scholarship.

INAUGURATION OF SSN IEEE SPS - 2023



The Inauguration of the IEEE Signal Processing Society, conducted on **August 21, 2023**, by the SPS student chapter at the ECE Seminar Hall, marked the beginning of the society's activities for the term. The event, attended by 123 participants from 3rd and 4th Year BME students, commenced at 10:00 AM. Key activities included the **swearing-in of newly elected leaders**, such as the Chair, Vice Chair, and core Committee members, who outlined their vision and goals for advancing the field and promoting research and education in signal processing.

Dr. M.D. Selvaraj, Professor, Dept. of ECE, IITDM Kancheepuram delivered an informative talk on the latest advancements in Visible Light Communication. The event also provided networking opportunities for professionals, researchers, and academics, fostering collaborations and the exchange of ideas. The inauguration concluded with a celebratory gathering, allowing attendees to socialize and connect informally.



TWO DAY WORKSHOP ON “MATLAB AND SIMULINK FOR ARTIFICIAL INTELLIGENCE WITH LOW-COST HARDWARE INTERFACING”

The workshop took place at **DSP Lab, Department of ECE** for a period of two days – **22/08/2023 and 23/08/2023**. It aimed to equip students with fundamental insights into the capabilities of **MATLAB and Simulink for AI and ML**. It was structured to provide a comprehensive understanding and hands-on experience on the **first day**, which primarily delved into the basic features of MATLAB and Artificial Intelligence. This included practical demonstrations using Machine Learning and Deep Learning toolboxes, in addition to introducing the participants to the available on-ramp courses in MATLAB.

On the **second day**, the focus shifted towards an overview of **hardware support from MATLAB and Simulink**. Participants were treated to a demonstration showcasing the interfacing capabilities of MATLAB and Simulink with devices like Arduino and Raspberry Pi. The hands-on sessions were particularly lauded by the attendees, who found them both informative and fascinating.

Hands-on demos on Machine Learning and Deep Learning toolboxes. Introduction to available MATLAB on-ramp courses, Demonstration of MATLAB and Simulink's interface with Arduino and Raspberry Pi. Informative and engaging hands-on sessions left the participants enthused and enlightened.

The workshop was a resounding success, offering a platform for budding engineers to explore the practical applications of MATLAB and Simulink in the realm of Artificial Intelligence while delving into low-cost hardware interfacing.



SPS SCHOLARSHIP PROGRAM

*-Supraja Vaidhyathan, BME 2020 - 24
Biomedical engg. Ph.D Student, University of Houston*

The application process for the **SPS Scholarship Program** opens in **March** every year and remains open until the end of **June**. Applications are submitted **through the IEEE website**. The primary eligibility criterion for this program is **membership in the IEEE Signal Processing Society**.



Under this program, a total of **\$7,000** is provided as a scholarship in three installments: \$2,000 in each of the first two years and \$3,000 in the final year. It is recommended to renew the application for the 2nd and 3rd years.

The application includes several questions, such as:

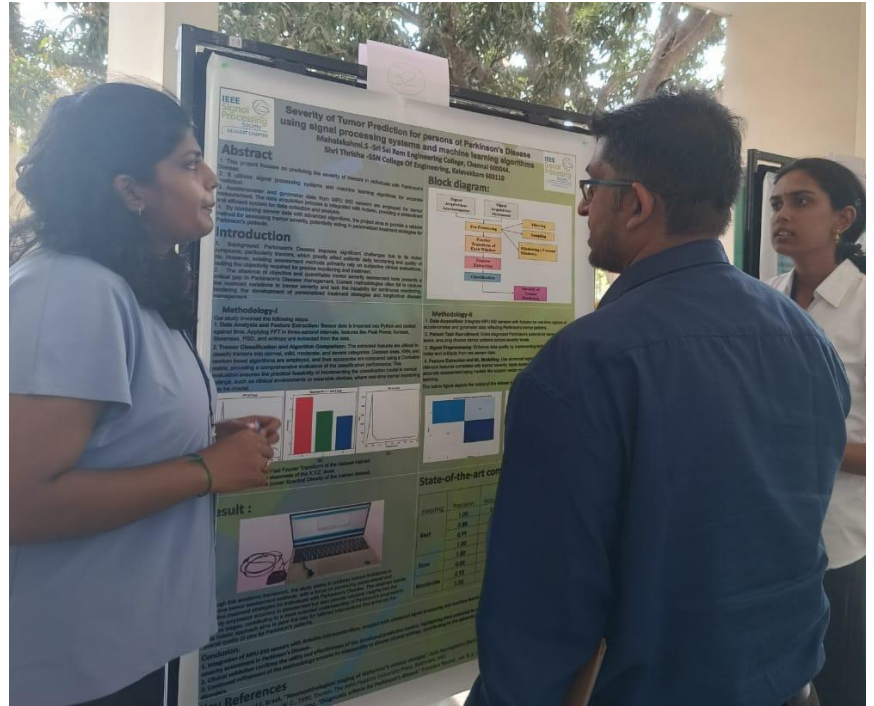
- The signal processing courses you have completed in past semesters or plan to complete in upcoming semesters (a minimum of 3 subjects).
- Projects and achievements you have accomplished.
- Hackathons and competitions you have participated in.
- The need for this scholarship.
- Why you believe you are eligible for the program.
- How you intend to use the scholarship.
- Where you see yourself in the next 5-10 years.
- Letters of Recommendation (LOR) from two of your project guides.

I received my first-year scholarship of \$2,000 and used it to complete my final year project at NIMHANS, Bangalore. Currently, I have applied for the renewal of the scholarship for the second year.

SYMPOSIUM ON PH.D FORUM

-Shri Thrisha Arunkumar, BME 2020 - 24

On March 1st and 2nd, PDEU, Gandhinagar, hosted a dynamic PhD Forum in collaboration with IEEE Signal Processing Society Gujarat Chapter. Gathering esteemed scholars, industry experts, and budding researchers, the event served as a melting pot of ideas, insights, and inspiration



Day one commenced with illuminating lectures, setting the tone for the forum's intellectual discourse. Dr. K S Sowmiya Ani, renowned for her expertise in AI Assisted Scientific Writing and Publishing, captivated the audience with her insights garnered from years of experience as the founder editor and reviewer at Sowmis_AWW, and as an editor at Cactus Communications. Her session not only illuminated the nuances of effective scientific communication but also underscored the pivotal role of technology in modern research dissemination.

Following this enlightening start, Dr. Sanjeev K Vafsrhney, a distinguished scientist at the Department of Science and Technology, Government of India, explained the importance of research and delineated various national and international opportunities for aspiring scholars. His address served as a compass, guiding young researchers through the myriads of academic pursuits.

SYMPOSIUM ON PH.D FORUM

-Shri Thrisha Arunkumar, BME 2020 - 24

Following this session we had a talk on “Navigating the journey from research to paper publication” by Dr. Dhaval Pujara, Director of SOT, PDEU. He gave insights on translating research findings into impactful publications, demystifying the process for aspiring scholars.



Day two focused on entrepreneurship opportunities in India, presented by Mr. Ravi Patel, and real-world case studies bridging academic research and industry success, led by Mr. Nilesh Ranpura. Attendees also had the chance to showcase their research projects, highlighting the diverse areas of study within the field.

At the end of the session budding researchers showcased their projects across various domains of Signal Processing. I had the opportunity to present a poster on the Severity of Tremors Prediction for Persons of Parkinson’s Diseases using Machine Learning and Signal Processing Algorithms. Though I was not able to win in my track I was awarded an Appreciation award.

The PhD Forum has been more than just an event; it has been a transformative experience, shaping the trajectory of my academic journey and instilling in me a lifelong commitment to the pursuit of knowledge and innovation.



ICASSP '24 - SEOUL

Dr. N. Venkateswaran, Professor, Dept. of ECE

The IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP) 2024, held from April 14 to 19, included a comprehensive program that highlighted advances in biomedical engineering and signal processing.



The conference featured a **series of plenary speeches** by notable professionals such as Seongwhan Lee, Bhaskar Rao, Daniel Lee, and Jitendra Malik. These sessions presented in-depth insights into the most recent research and technology advances in signal processing, as well as their applications in biomedical engineering.

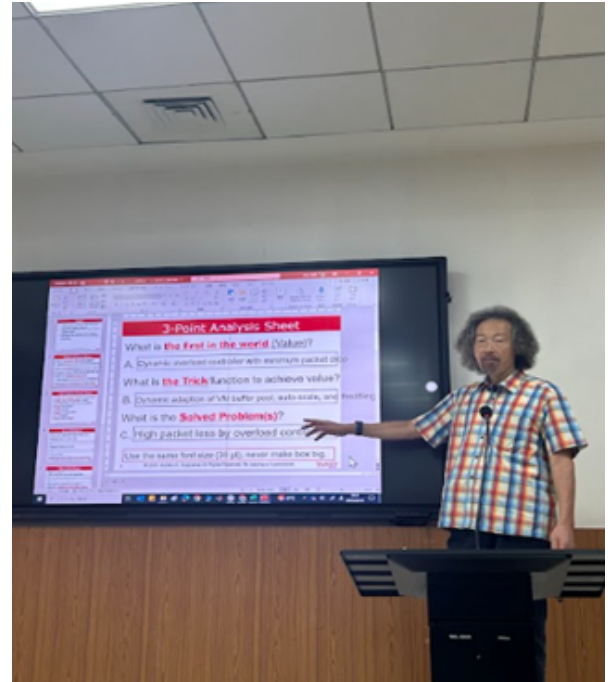
In addition to plenary sessions, the conference featured a **number of workshops and tutorials on biomedical signal processing**. These workshops offered hands-on learning opportunities and in-depth talks about medical imaging, health informatics, and machine learning applications in healthcare.



The **industry workshops**, which included leading organizations such as **MathWorks** and **Meta**, enhanced the curriculum by demonstrating realistic implementations and real-world applications of signal processing technology in the biomedical area.

SHORT COURSE ON “EASY AND SOLID LOGICS FOR RESEARCH PAPER WRITING IN HIGH IMPACT JOURNALS”

Dr. Akihiko (Ken) Sugiyama, currently with **Yahoo! JAPAN Research**, is a prominent figure in signal processing, particularly in audio coding and noise control. His distinguished career includes a lengthy tenure as a **research engineer at NEC Central Research Laboratories** and various key roles in the IEEE Signal Processing Society (SPS).



Day 1 began with a slide presentation focused on a three-point analysis, which was elucidated using various examples and demonstrations. Following a tea break, participants were tasked with assignments. They were instructed to independently choose a research paper, analyse its contents, and complete the assignment that consisted of Value, Trick and Problem, thus effectively evaluating the chosen paper. Dr. Sugiyama later reviewed these assignments, addressing any doubts and offering fresh insights and ideas.

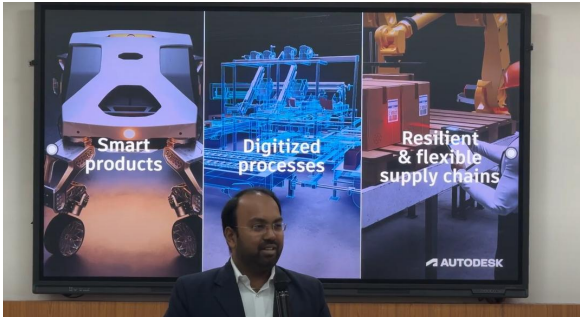
SHORT COURSE ON “EASY AND SOLID LOGICS FOR RESEARCH PAPER WRITING IN HIGH IMPACT JOURNALS”

On **Day 2**, the session started with an introduction to the principles of graph design and valuable tips for effective drafting. The participants were then assigned a fresh set of assignments aimed at testing their comprehension and application of the newly acquired knowledge. Additionally, the significance of crafting a well-structured abstract was emphasized, and this critical skill was revisited and reviewed later in the session to ensure its importance was thoroughly understood by all participants.

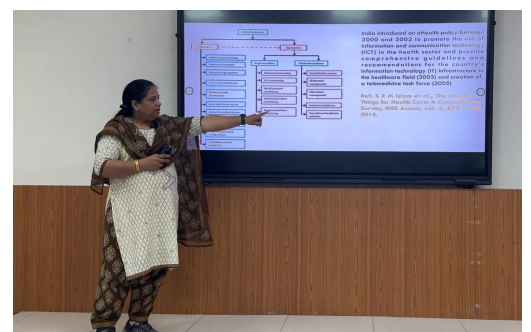


The short course also had a Distinguished Lecture (DL) on "Personal Information Devices". The DL had over 60 participants from various field of interest. The talk was truly captivating and offered a profound insight into the fascinating history of various devices. It skilfully unravelled the evolution of these devices, providing a comprehensive perspective on how they have transformed over time. Various technologies such as Audio encoding and decoding were explained to the participants.

SEASONAL SCHOOL PROGRAM ON "AI BASED MEDICAL SYSTEMS DESIGN FOR BIO SIGNALS"



On the **first day**, the speakers included : Dr. S. Kanagaraj, Professor, Department of Mechanical Engineering, IIT Guwahati; Dr. Sainarayanan Gopalakrishnan, HCL Technologies, Chennai; Dr. B. Geethanjali, Associate Professor, Department of Biomedical Engineering, SSN College of Engineering, Chennai; and Dr. S. Pravin Kumar, Associate Professor, the Department of Biomedical Engineering, SSN College of Engineering, Chennai.



On the **second day**, the speakers included: Dr. Rajeshwari Aghoram, Additional Professor, Department of Neurology, JIPMER, Pondicherry; Dr. Vivek Maik, Principal Project Scientist, IIT Madras; and Mr. S. Afsal from MathWorks.

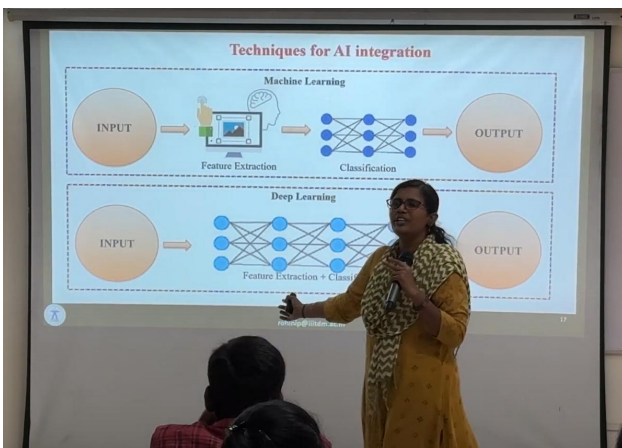
On the **third day**, the speakers included: Dr. R. Periyasamy, Assistant Professor, Department of Instrumentation and Control Engineering, NIT Trichy; Dr. M. Dhanalakshmi, Assistant Professor, Department of Biomedical Engineering, SSN College of Engineering, Chennai; and Mr. Kanagaraj K. from AD Instruments.

SEASONAL SCHOOL PROGRAM ON "AI BASED MEDICAL SYSTEMS DESIGN FOR BIO SIGNALS"



On the **fourth day**, the speakers included: Dr. Priyanka Kokil, Associate Professor, Department of Electronics and Communication Engineering, IIITDM Kancheepuram; Dr. Vinayak from Speed Institute, Chennai; Dr. Ilankumaran Kaliamoorthy from Rela Hospitals, Chennai; and Dr. V. Prasath from AutoDesk, Bangalore.

On the **final day**, the speakers included: Dr. Bibin Francis, Assistant Professor, Department of Electronics and Communication, NIT Trichy; Dr. P. Rohini, Assistant Professor, Department of Electronics and Communication, IIITDM Kancheepuram; and Dr. A. Balaji Ganesh, Dean (R&D), Velammal Engineering College, Chennai.



The program provided a valuable platform for participants to gain cutting-edge knowledge and practical insights from experts in academia, industry, and healthcare. The diverse range of topics covered by the 17 distinguished speakers facilitated a comprehensive understanding of AI applications in medical system design. The event fostered a vibrant exchange of ideas and experiences, laying the groundwork for future innovations and research in the field.

Overall, the seasonal school was a resounding success, inspiring participants to further explore and contribute to the advancements in bio-signal processing and AI-based medical technologies.

SPS DAY TALK



The **IEEE SPS Day- Deep Oscillatory Neural Networks For Signal Processing** was conducted on **24th May 2024** by the SPS student chapter at the ECE seminar hall. There were 53 participants from BME, ECE department. The event started at 10:00 AM. **Dr.V.Srinivasa Chakravarthy, Professor, Dept. of Biotechnology, IIT Madras** delved into a cutting-edge intersection of neuroscience-inspired algorithms and signal analysis. Then we were briefed about the advantages of DONNs over traditional neural networks, highlighting their robustness to noisy and non-stationary signals, as well as their potential for learning complex temporal patterns. He also gave a novel approach to processing time-series data, particularly in fields like speech recognition, image processing and biomedical signal analysis. He explained how DONNs utilize deep learning techniques to extract hierarchical features from signals, while also leveraging oscillatory dynamics to enhance model performance. The session was informative and fun and the professor taught good and at a pace we could all keep up and understand.



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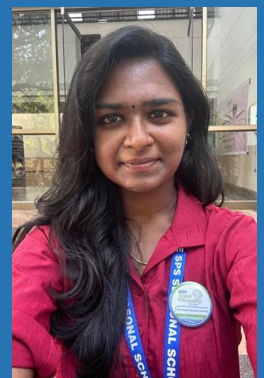
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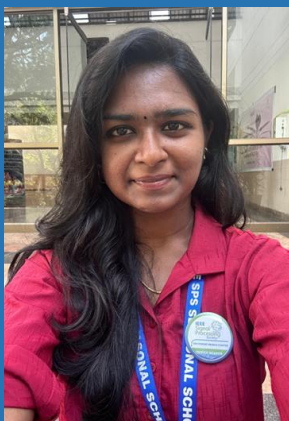
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